Eloy Mosig García, Ph.D. Student in Mathematics

Email me GitHub +34 689 80 19 67
Personal website

Ph.D. Student in Mathematics at Università di Pisa. My main research interests are statistical learning theory and topological data analysis.

Employment and education history

2026 – present ◇ **Research Fellow (Incarico di Ricerca), Università di Pisa** in the project **OTODDLE** – Optimal Transport and Large Deviations for Deep Learning Theory, FIS 3 Consolidator Grant funded by the Italian Ministry of University and Research (MUR), PI Prof. Dario Trevisan. Research on quantitative limit theorems and optimal transport methods for deep learning theory.

2023 – 2026 ◇ **Ph.D., Università di Pisa** Mathematics.
On Sep 2026 I was invited by Professor E. Di Bernardino to do a research visit at **Laboratoire J. A. Dieudonné, Université Côte d'Azur**. During Apr 2025 – Jun 2025 and Sep 2025 – Dic 2025 I was invited by **IMSV, Universität Bern** to do a research visit with Professor A. Agazzi. During Jan 2026 – Aug 2026 I did an intership at **Qi4M, Rome**. I expect to discuss my thesis in summer 2027.

2021 – 2023 ◇ **Master's Degree, Università di Bologna** Mathematics.
Thesis title: *Some new results on the matching distance in biparameter persistent homology*.

2019 – 2021 ◇ **Bachelor's Degree, Universidad Complutense de Madrid** Mathematics.
Thesis title: *Results in persistent homology and application to the neo-riemannian Tonnetz*.

Research Publications

1. A. Agazzi, G. Bruno, E. Mosig García, S. Saviozzi, and M. Romito, *Stochastic scaling limits and synchronization by noise in deep transformer models*, 2026. arXiv: 2604.26898 [math.PR]. URL: https://arxiv.org/abs/2604.26898
2. A. Agazzi, E. Mosig García, and D. Trevisan, *Quantitative gaussian-process limits of tensor programs*, 2026. arXiv: 2607.06290 [cs.LG]. URL: https://arxiv.org/abs/2607.06290
3. F. Conti et al., *The convex matching distance in multiparameter persistence*, 2026. arXiv: 2512.02944 [math.AT]. URL: https://arxiv.org/abs/2512.02944
4. P. Frosini, E. Mosig García, N. Quercioli, and F. Tombari, "Matching distance via the extended pareto grid," *SIAM Journal on Applied Algebra and Geometry*, vol. 9, no. 3, pp. 554–576, 2025. DOI: 10.1137/24M1680398 eprint: https://doi.org/10.1137/24M1680398.
5. E. Mosig García, A. Agazzi, and D. Trevisan, "Quantitative convergence of trained neural networks to gaussian processes," in *Advances in Neural Information Processing Systems*, D. Belgrave et al., Eds., vol. 38, Main Conference, Curran Associates, Inc., 2025, pp. 164 679–164 726. DOI: 10.52202/085713-5492

Talks

Jun 2026 ◇ **5th Italian meeting on Probability and Mathematical statistics, Palermo** Quantitative Master Theorems for Tensor Programs via the Wasserstein distance

Mar 2026 ◇ **Topology seminar, CUNEF (Madrid)** The Pareto grid in multiparametric persistence

Talks (continued)

Feb 2026 ◇ **TDA days, Universidad de Sevilla** The convex matching distance in multiparametric persistence

Jan 2026 ◇ **Mathematics for AI and ML, UMI Workshop, Università di Roma Sapienza** Quantitative convergence of trained single layer deep neural networks to Gaussian processes.

Oct 2025 ◇ **CVG Seminar, Bern** Quantitative convergence of trained single layer deep neural networks to Gaussian processes.

Sep 2025 ◇ **Is AI Intelligence? A Dialogue Between Neuroscience and Physics, Università di Roma Sapienza** Neural Manifold Learning through AI.

Jan 2025 ◇ **IMSV Retreat, Münchenwiler** Quantitative convergence of trained deep neural networks to Gaussian processes.

◇ **Gazteak RSME 2025, Universidad de Bilbao** The extended Pareto grid in multiparameter persistent homology.

Jul 2024 ◇ **9th European Congress of Mathematics, Sevilla** Computation of the 2-parameter matching distance via the extended Pareto grid.

Nov 2023 ◇ **ICMAT, Madrid** A Morse Theory approach to the computation of the matching distance.

Poster sessions

Dec 2025 ◇ **NeurIPS 2025, San Diego** Quantitative convergence of trained single layer deep neural networks to Gaussian processes.

Aug 2024 ◇ **SPIRES, University of Oxford** Matching distance via the extended Pareto grid.

Feb 2024 ◇ **Universitat de Barcelona** Extended Pareto grid in multiparameter persistent homology.

Nov 2023 ◇ **AATRN, 9th Mexican Workshop on Applied Geometry and Topology** Computing biparameter matching distance via the extended Pareto grid.

Organization of academic events

Jun 2026 ◇ **PiBe de oro workshop, Pisa** Pisa/Bern meeting on Probability and Statistics.

Attendance to academic events

Sep 2025 ◇ **Centro di Ricerca Ennio di Giorgi, Pisa** Workshop in combinatorial algebraic topology and applications III.

May 2025 ◇ **Centre de recherches Mathématiques, Montréal** Mathematical foundations of data science.

◇ **Isaac Newton Institute, Cambridge** Uncertainty in multivariate, non-Euclidean, and functional spaces: theory and practice.

Jan 2025 ◇ **Bari, Italy** UMI Workshop: Mathematics for AI and machine learning.

Dic 2024 ◇ **Centro di Ricerca Ennio di Giorgi, Pisa** Optimal transportation and applications.

Nov 2024 ◇ **Centro di Ricerca Ennio di Giorgi, Pisa** Workshop in combinatorial algebraic topology and applications II.

Sep 2024 ◇ **Cetraro, Italy** CIME summerschool: High-dimensional approximation: from theoretical foundations to machine learning and PDEs.

◇ **Cortona, Italy** Mathematics of machine learning in Cortona.

Jun 2024 ◇ **Rome** 4th Italian meeting on probability and mathematical statistics.

Jan 2024 ◇ **Università Bocconi, Milan** UMI Workshop: Mathematics for AI and machine learning.

Attendance to academic events (continued)

Nov 2023 ◇ **Centro di Ricerca Ennio di Giorgi, Pisa** Workshop in combinatorial algebraic topology and applications I.

Sep 2023 ◇ **Universitatea din Bucuresti** Moduli and Friends conference.

Theses supervised

I supervised together with Prof. A. Agazzi the master thesis of Karl-Joan Alesma at University of Bern.

Skills

Languages	◇ English (C1), Italian (C1), French (A2), Spanish (native).
Programming languages	◇ Python, C++, Java, HTML.
Mathematical software	◇ SageMath, MatLab.
Document preparation	◇ LaTeX, Microsoft Office Package, Libre Office Package.

Employment History

Jan 2026 – Aug 2026 ◇ **Qi4M, Rome, Internship** Fine-tuning of visual language models for table recognition.

Feb 2024 - Jun 2024 ◇ **Università di Pisa, Teaching Assistant** Teaching Assistant for the course of "Mathematical Statistics" for Mathematics Bachelor students.

Nov 2023 - Dic 2023 ◇ **Online** Support lessons to Mathematics Bachelor students.

Nov 2022 - Apr 2023 ◇ **Università di Bologna, Internship** Programming of a Python-based visualisation tool to be used in research on persistent homology.

Jun 2018 - Sep 2018 ◇ **UNED, Madrid, Internship** Programming and design of a telegram bot managed by UNED Abierta to be used in the prevention of gender-based violence.

Personal interests

◇ I am a piano player, and during my university years, I participated in some music groups playing diverse genres, from jazz and funk to classical music. I am also a passionate reader and during my bachelor years organised weekly informal literature encounters and debates in Madrid.

Awards and Achievements

2020 ◇ Publication of two short stories, under La Gota de Leche.

2015 ◇ **II Philosophy Olympiad of Spain, First Award**, Dissertation published under Ediciones La Torre.